

COFFEE PROCESSING & EXPORT MANAGEMENT

From cherry to shipment, in one system.

MADDA is an ERP built specifically for Ethiopian coffee exporters. It follows a lot of coffee through every hand it touches — from the farmer at the washing station, through processing and the warehouse, into a proforma invoice, onto a container ship, and finally into the bank. Everything that today lives in one fragile spreadsheet becomes a secure, multi-user, offline-ready system.

The problem with the spreadsheet

Today the whole operation is tracked in a single Excel workbook: cherry purchases, processing batches, inventory, sales and payments. It works — until it doesn't. One mistyped ID breaks traceability. Anyone with the file can edit anything. There is no history of who changed what. And when two people fill it in at once, one of them loses their work.

No access control

Every user sees and edits every cell. No roles, no audit trail.

Manual traceability

Lot IDs are typed by hand. A single error severs the farm-to-buyer chain.

No documents

Proforma and commercial invoices are made outside the system, with no numbering rules.

MADDA replaces it with one source of truth — role-based, auditable, mobile-first, and designed to keep working in places with no internet.

01 — THE LIFECYCLE

One documented path from sourcing to settlement

Every step below is a module in the system. Data flows automatically from one to the next — a purchase becomes a processing batch, which becomes a lot, which becomes inventory, which becomes a sale.



01 · Sourcing & suppliers

Farmer, cooperative and collector profiles with contact, location and bank details. Buyers get their own customer profile.

02 · Cherry purchasing

Each delivery is recorded at the station: supplier, weight in KG, price per KG, receipt number, total and payment status. The system prints a receipt and numbers it automatically.

03 · Processing

Cherry is processed into parchment, dry parchment and green bean. MADDA records moisture, grade, screen size and cupping score, calculates the **yield %** automatically, and creates a traceable **Lot ID**.

04 · Inventory & traceability

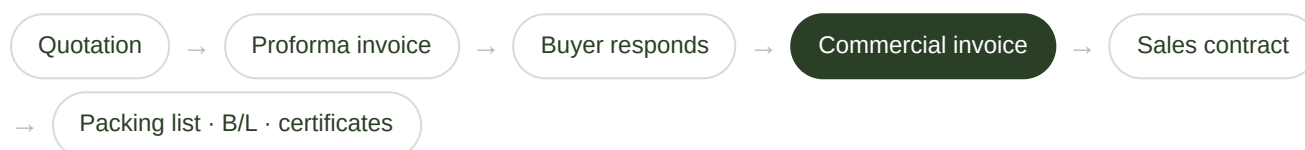
Green coffee enters inventory per lot and warehouse with unit cost and valuation. The full chain — purchase → batch → lot → inventory → sale — is preserved, so any bag can be traced back to the farm.

02 — SELLING, SHIPPING & MONEY

From a first quotation to money in the bank

05 · Sales & export — the document pipeline

This is where MADDA replaces email threads and ad-hoc invoices with a controlled, numbered pipeline. A quotation becomes a **proforma invoice**; when the buyer responds, the proforma moves forward step by step and is **converted into a commercial invoice that keeps its reference number**.



Each document keeps its own number series (PRO-2026-0001, COM-2026-0001) and an activity history showing every status change and every email sent to the buyer.

06 · Shipping & tracking

Every consignment gets a tracking ID and a simple timeline: **Received** → **In Transit** → **Delivered**, updated from the phone. A built-in checklist tracks the full export document pack — CLU quality certificate, phytosanitary, ICO and chamber certificates of origin, bill of lading, bank permit, insurance and more.

08 · Finance

Purchases, expenses and payments in multiple currencies (ETB, USD, EUR, GBP) with stored exchange rates. Accounts-payable and accounts-receivable ageing show who is owed and who owes. Profitability can be viewed by station, lot and buyer.

07 · Contracts

Sales contracts record Incoterms, value, currency and the ECTA registration reference, linked to the proforma and the shipment.

Approvals

Off by default. When enabled, purchases, expenses or payments above a threshold raise an approval request that appears in a queue — with a full decision trail.

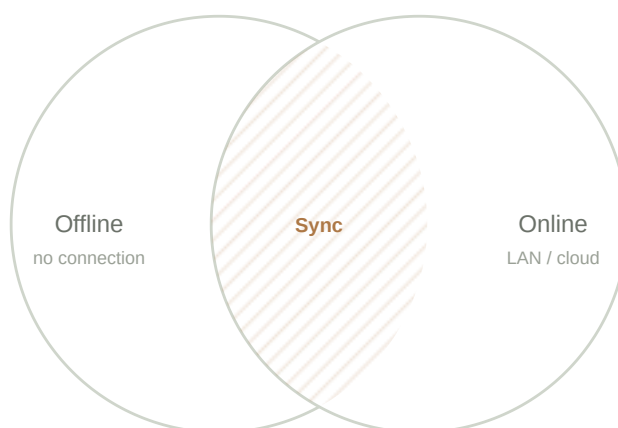
Reports & the executive dashboard

The CEO dashboard shows live figures with no spreadsheet formulas: active stations, cherry purchased, green output, inventory, purchase cost, expenses, sales revenue and gross profit — plus station-level performance, expense breakdowns by category, and green output by process and grade.

03 — WHERE IT RUNS

Works offline. Speaks Afaan Oromoo.

Washing stations often have no internet. MADDA is a phone-first application that keeps working without a connection: staff record a full day of deliveries, batches and expenses on the device, and everything uploads automatically the moment connectivity returns.



Phone-first design

Built for a medium phone screen. Tables become readable cards, navigation becomes a bottom bar, and every action is reachable with a thumb.

Two languages

The entire interface is available in **English** and **Afaan Oromoo**, with reference data (processes, grades, categories) translated too.

Email built in

Proforma and commercial invoices can be emailed to the buyer from inside the system, and every sent email is logged on the record so you can see exactly what was sent and when.

Audit & history

Every change is recorded — who, what, when. Nothing is hard-deleted. Full history on every invoice, shipment and customer.

04 — WHO USES IT

Everyone sees exactly what they should

Access is role-based and deny-by-default. A warehouse officer can update stock but cannot see prices or create purchases. A station manager works only with their own station. The CEO sees everything.

ROLE	WHAT THEY DO IN MADDA
CEO / Owner	Full visibility, executive dashboard, high-value approvals.
General Manager	Operational oversight, approvals, all reports.
Finance Manager / Accountant	Payments, expenses, AR/AP ageing, currency, profitability.
Sales / Export Manager	Buyers, quotations, proforma and commercial invoices, contracts.
Export Documentation Officer	Export document checklist, certificates, customs and bank paperwork.
Procurement Manager	Suppliers, cherry purchase approvals and pricing.
Station Manager	Runs a station: cherry intake, processing, local expenses.
Cherry Receiver	Registers deliveries, weighs, records the farmer, prints a receipt.
Processing Supervisor	Records processing batches and conversions.
Quality / Cupping Officer	Grades, moisture, cupping scores and quality data.
Warehouse Officer	Stock in/out, locations, counts and lot status.
Logistics Officer	Shipments, tracking, containers and transport status.
Auditor	Read-only access to every record and the audit trail.

05 — UNDER THE HOOD

Modern, local-first, and yours to host

Backend & data. A Node.js API with a PostgreSQL database — the same technology used by banks and logistics companies. The data model covers the full coffee chain, not just the dashboard.

Frontend. A React application installed on a phone like an app (PWA), with local storage so it works offline and syncs in the background.

Deployment. Runs entirely on a local machine or office server today, with no cloud dependency. It can move to a hosted server later without changing how people use it.

Extensible. Currency, numbering rules, approval thresholds and reference lists are all configurable in settings — no code changes needed.

THE OUTCOME

One system that runs the whole export business — and never loses a bag of coffee along the way.

Traceability buyers trust

Every lot carries its full story from farmer to container — increasingly a requirement for international buyers and export markets.

No more spreadsheet risk

Role-based access, full audit trail and no lost edits. The business stops depending on one file and one person.

Faster, cleaner export paperwork

Numbered proforma and commercial invoices, a document checklist per shipment, and emails logged automatically.

Decisions from live numbers

Real-time dashboard and reports instead of formulas — cost, yield, revenue and profit by station, lot and buyer.

Built for the reality of the field

Offline-first on phones, in Afaan Oromoo and English, so the people at the washing station can actually use it.

Ready to grow

A solid data foundation for future needs — certifications, full accounting, EUDR geolocation, market-price tracking.

MADDA

Coffee Processing & Export ERP — built for Ancient Halo Coffee Export.